

Software Requirements Specification (SRS)

Project Name: SmartStream – Integrated Business Management System

Client: KTD LANKA (PVT) LTD

Group No: 07 - NextGen Developers

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1. Introduction

1.1 Purpose and Scope

The purpose of this document is to define the requirements for "SmartStream," a cloud-based, AI-enhanced business management system developed for KTD LANKA (PVT) LTD.

The scope of this system includes smart inventory management, AI-powered invoice processing via OCR, distributor GPS tracking, automated email/SMS alerts, and role-based access control. The system aims to replace manual, paper-based workflows with a centralized platform to improve operational efficiency, accuracy, and executive decision-making.

1.2 Definitions, Acronyms, Abbreviations

- **SRS:** Software Requirements Specification
- **OCR:** Optical Character Recognition
- **AI:** Artificial Intelligence
- **RBAC:** Role-Based Access Control
- **GPS:** Global Positioning System
- **Cloud System:** A network of remote servers hosted on the internet to store, manage, and process data.

1.3 References

- Initial Client Requirement Communications (WhatsApp Records from KTD LANKA)
- SENG 31242 System Design Project Guidelines
- Fact-Finding Session Documentation (Interview Transcripts, Survey Data)

1.4 Overview of Document

This document is organized into six main sections. Section 2 provides a high-level overview of the system and its users. Section 3 details the analysis phase, including fact-finding and use cases. Sections 4 and 5 specify the functional and non-functional requirements. Section 6 concludes with a feasibility study of alternative solutions.

2. Overall Description

2.1 Product Perspective

SmartStream is a new, independent software solution designed to act as the central nervous system for KTD LANKA's distribution and inventory operations. It will interface with cloud storage systems for disaster recovery, email/SMS gateways for alerts, and utilize mobile device GPS hardware for tracking distributors.

2.2 User Classes and Characteristics

- **Director:** Requires a high-level view of operations, easy-to-read dashboards, financial reports, and remote access. Not highly involved in technical daily data entry.
- **Stock Manager / Inventory Staff:** Responsible for maintaining stock levels. Needs rapid data entry, barcode/OCR support, and low-stock alerts.
- **Accountant:** Needs accurate financial data, automated invoice processing, and seamless integration with payment tracking.
- **Distributor:** Field workers using mobile devices. Require a simplified UI for mobile, GPS tracking capabilities, and easy photo upload features for invoices.
- **System Admin:** Responsible for managing roles, access control, and cloud backups.

2.2.1 User Personas (Detailed)

1. The Director (Executive Management) - Mr. K.T. Dias

Demographics: 52 years old, frequently travels for business, relies heavily on his iPad and laptop.

Goals: Wants real-time visibility into overall company performance, total inventory value, and live distributor locations. Needs to make quick, data-driven financial decisions.

Pain Points & Frustrations: Highly frustrated by the current manual system where billing delays occur because of lost paper invoices. Hates waiting for end-of-week manual reports to know how the business is performing.

2. The Distributor (Field Operations) - Kamal Fernando

Demographics: 29 years old, works out in the field 90% of the day. Uses a mid-range Android smartphone.

Goals: Wants to complete his delivery routes quickly without being slowed down by complicated paperwork. Needs a mobile app that is fast, easy to navigate with one hand, and doesn't drain its battery.

Pain Points & Frustrations: Currently frustrated by the responsibility of keeping track of physical paper invoices, which frequently get lost or damaged in transit. Dislikes typing long customer or product details on a small mobile screen.

3. The Stock Manager (Warehouse Operations) - Nuwan Perera

Demographics: 35 years old, stationed at the main KTD Lanka warehouse. Uses a desktop computer and a tablet.

Goals: Needs to ensure the warehouse never runs out of popular items. Wants the system to automatically notify him via SMS/Email when supplies drop below a certain threshold, so he doesn't have to check manually.

Pain Points & Frustrations: Currently struggles with the static, manual Excel-based tracking system. Extremely frustrated by the constant discrepancies between what the Excel sheet says and the actual physical stock on the shelves.

4. The accountant (Finance Department) - Sanduni Silva

Demographics: 31 years old, works in the main office, and is highly proficient in Excel and accounting software.

Goals: Wants to eliminate manual data entry. Hopes the new AI OCR system will automatically extract product names and totals from distributor uploads so she can approve them with one click.

Pain Points & Frustrations: Wastes hours every week manually typing data from crumpled, hard-to-read paper invoices into the system. Frustrated by billing delays caused when distributors forget to hand in their delivery slips on time.

5. The System Admin (IT Support) - Shanuka Wijesinghe

Demographics: 27 years old, tech-savvy, works from a dual-monitor desktop setup.

Goals: Wants a reliable, automated system. Needs to easily assign different permission levels (e.g., ensuring a Distributor cannot access the Director's financial dashboard). Wants peace of mind that the cloud disaster recovery backups are running successfully every 12 hours.

Pain Points & Frustrations: Dislikes systems that are hard to maintain or require system downtime for minor updates. Frustrated when users share passwords, making strict role-based access control (RBAC) difficult to enforce.

2.3 Operating Environment

- **Web Application:** Accessible via modern web browsers (Chrome, Safari, Edge) on desktop and tablet.
- **Backend & Database:** Hosted on a secure cloud infrastructure (e.g., AWS, Azure) with high availability.

2.4 Design and Implementation Constraints

- **Mobile Connectivity:** Distributors may face varying internet connectivity in remote areas; the mobile app must handle delayed sync for uploads.
- **Security:** As it handles financial data, strict role-based access and data encryption are mandatory.

2.5 Assumptions and Dependencies

- The client will provide a stable internet connection for office staff.
- Distributors will have access to smartphones with functioning cameras and GPS capabilities.
- Third-party APIs (SMS gateway, Email service) will maintain high up time.

3. System Analysis

3.1 Fact-Gathering Techniques Used

- **Structured Interview:** Conducted with the Director and Accountant to understand

high-level business goals, current accounting pain points, and reporting requirements. (*Evidence: Interview notes and WhatsApp client request text*).

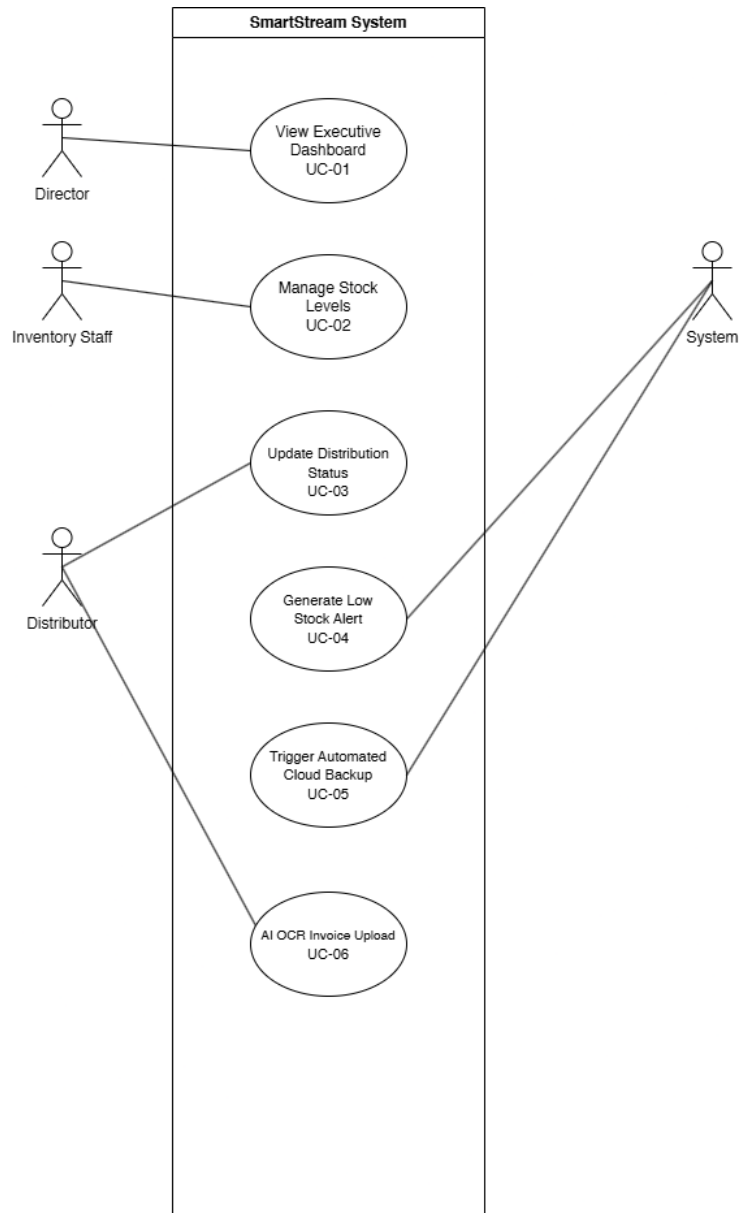
- **Questionnaire/Survey:** Distributed to Inventory Staff and Distributors to assess day-to-day workflow bottlenecks, manual data entry times, and comfort with mobile technology. (*Evidence: Anonymized responses and statistical summary*).

3.2 Existing System Analysis & Pain Points

KTD LANKA currently uses a manual, Excel-based system for inventory and paper invoices for distribution.

- **Pain Points:** Paper invoices are frequently lost, leading to delays in billing. Excel inventory tracking is static, leading to discrepancies between physical stock and recorded stock. Management lacks real-time visibility in company performance and distributor locations.

3.3 Use Case Diagram



3.4 Use Case Descriptions

UC-01: View Executive Dashboard

Field	Content
Use Case ID	UC-01
Use Case Name	View Executive Dashboard
Actor(s)	Director
Preconditions	The director is successfully logged into the web application.
Postconditions – Success	The system successfully retrieves and displays real-time key performance indicators, financial data, and inventory metrics.
Postconditions – Failure	System fails to retrieve real-time data; an error message or cached offline data is displayed.
Main Flow	1. The director navigates to the "Dashboard" module. 2. System verifies role-based access permissions. 3. System queries the cloud database for real-time sales/inventory data. 4. The system processes data into charts. 5. System renders interactive dashboard.
Alternative Flow 1	At step 3: If the cloud database is unreachable, the system loads offline cached data and displays a warning.
Business Rules	BR-01: Only users with the "Director" role can access financial totals.

UC-02: Generate Low Stock Alert

Field	Content
Use Case ID	UC-02
Use Case Name	Generate Low Stock Alert
Actor(s)	System (Trigger), Stock Manager
Preconditions	The system database has just processed an inventory deduction.
Postconditions – Success	An SMS and email alert is successfully delivered to the Stock Manager.
Postconditions – Failure	API fails to send the message; alert queued for retry.
Main Flow	1. Inventory deduction occurs. 2. The system checks new stock against threshold. 3. Detects stock is below threshold. 4. Retrieves Manager contact details. 5. Triggers Email/SMS API. 6. Manager receives notification.
Alternative Flow 1	If SMS Gateway is down, fallback to Email-only alert.
Business Rules	BR-02: Threshold levels must be manually configurable per product. BR-03: No duplicate alerts for the same product within 24 hours.

UC-03: Track Distributor Location

Field	Content
Use Case ID	UC-03
Use Case Name	Track Distributor Location
Actor(s)	Distributor, Director (Viewer)
Preconditions	Distributor logged in, GPS enabled, active route started.
Postconditions – Success	The system continuously logs coordinates, updating live maps.
Main Flow	1. Distributor selects "Start Route". 2. App locks onto GPS signal. 3. The app transmits coordinates to the backend every 60s. 4. Backend saves timestamped coordinates. 5. Director views updated location on map.
Business Rules	BR-04: Tracking terminates when route ends.

UC-04: Manage Inventory Levels

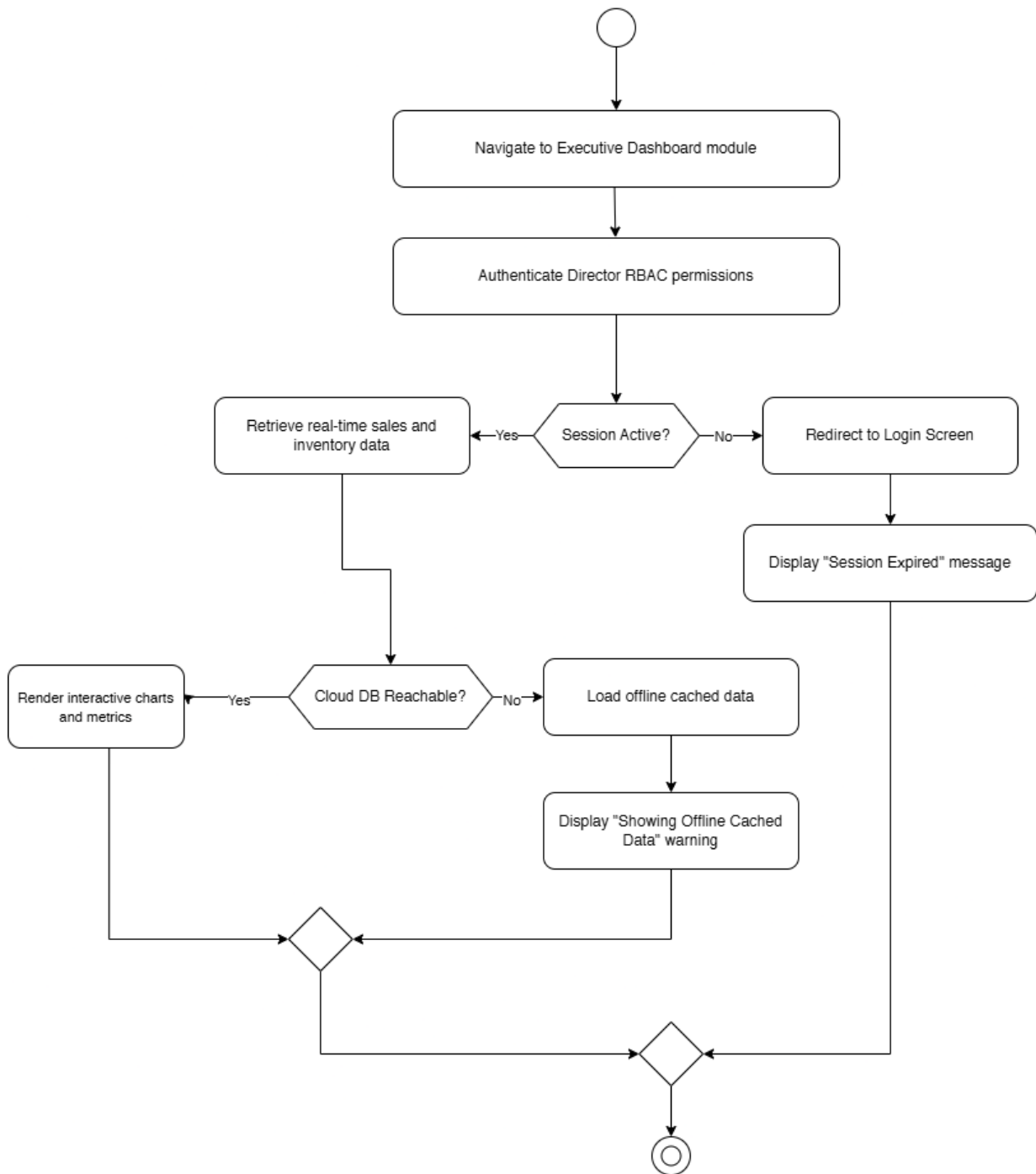
Field	Content
Use Case ID	UC-04
Use Case Name	Manage Inventory Levels
Actor(s)	Stock Manager, Inventory Staff
Preconditions	User logged in with Inventory Management permissions.
Postconditions – Success	Database updated with new count, system logs change.
Main Flow	1. The manager is searching for a product. 2. The manager inputs new physical quantities. 3. The system validates data. 4. System updates database. 5. Success confirmation displayed.
Alternative Flow 1	If negative input, reject and prompt correction.
Business Rules	BR-05: Manual adjustments must log timestamp and User ID.

UC-05: AI Invoice Image Upload and Processing

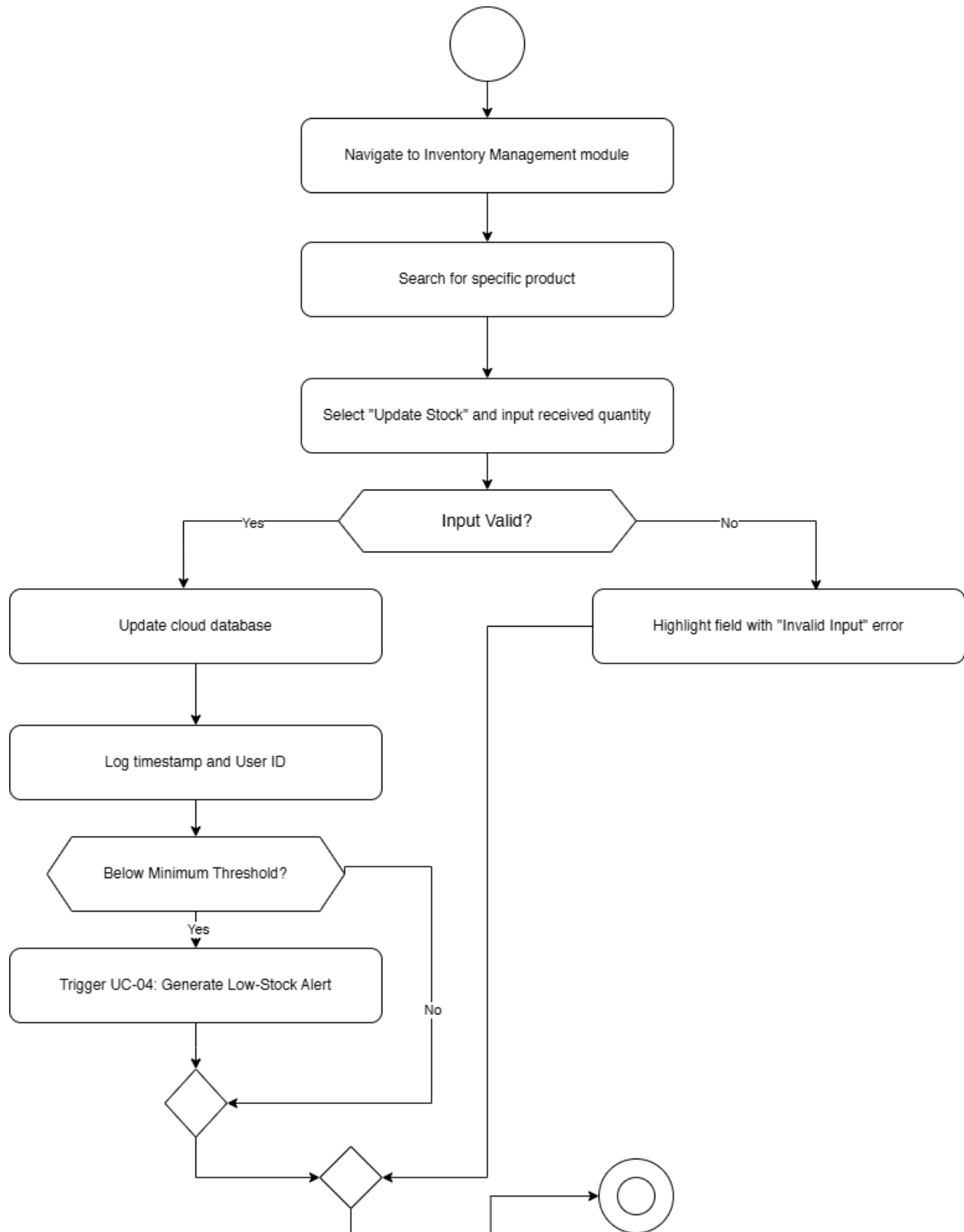
Field	Content
Use Case ID	UC-05
Use Case Name	AI Invoice Image Upload and Processing
Actor(s)	Distributor, AI OCR System, Inventory System
Preconditions	The distributor has a physical invoice and is logged into the app.
Postconditions – Success	Image saved, text extracted, inventory updated.
Main Flow	1. Distributor selects "Upload Delivery Invoice". 2. Camera activated, image captured. 3. AI OCR scans and extracts data. 4. System validates against the database. 5. Inventory updated. 6. Success alert sent.
Business Rules	BR-10: Invoices must have a timestamp/GPS. BR-11: OCR confidence >85% for auto-update.

3.5 Activity Diagrams

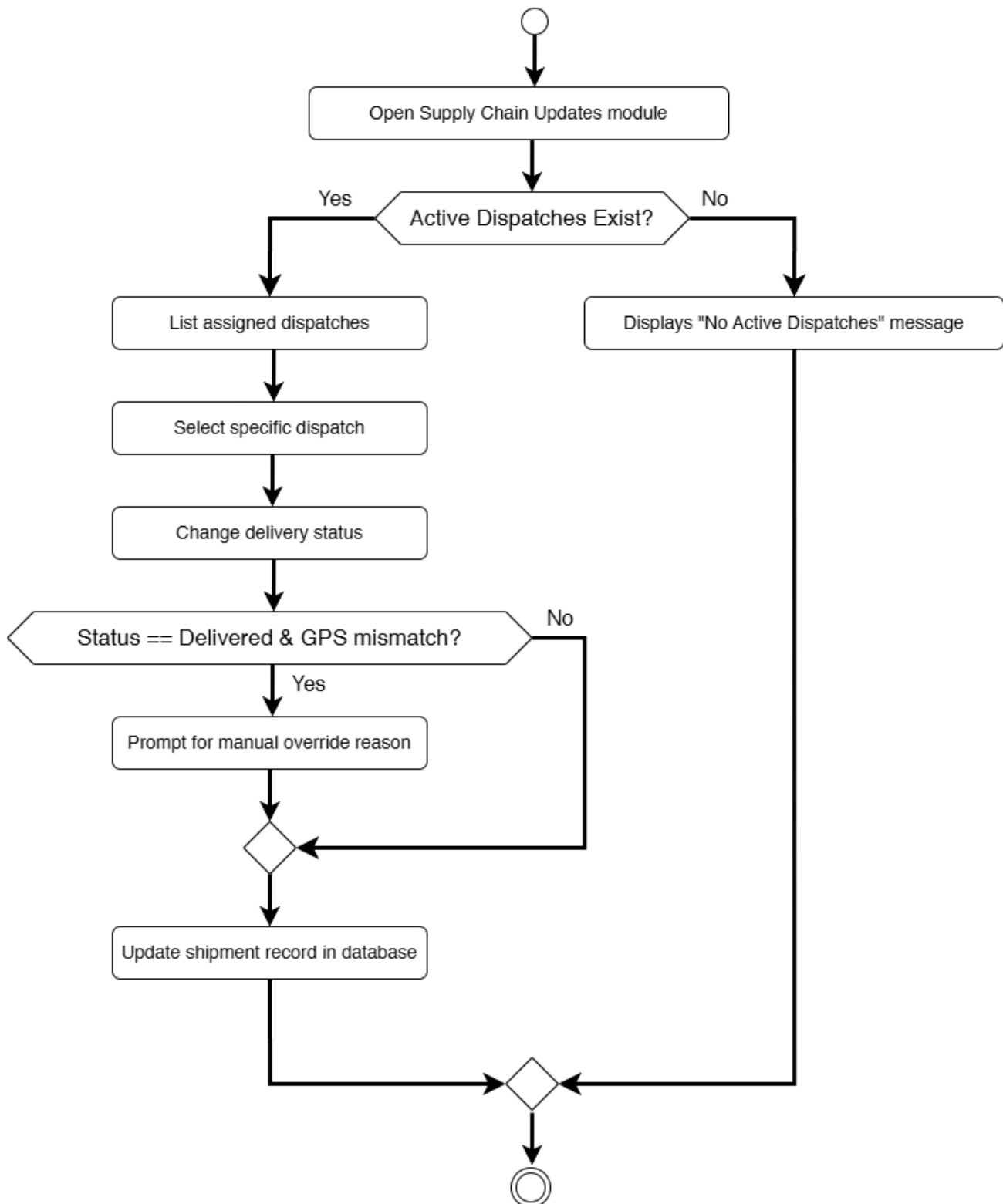
Activity Diagram: UC-01 View Executive Dashboard



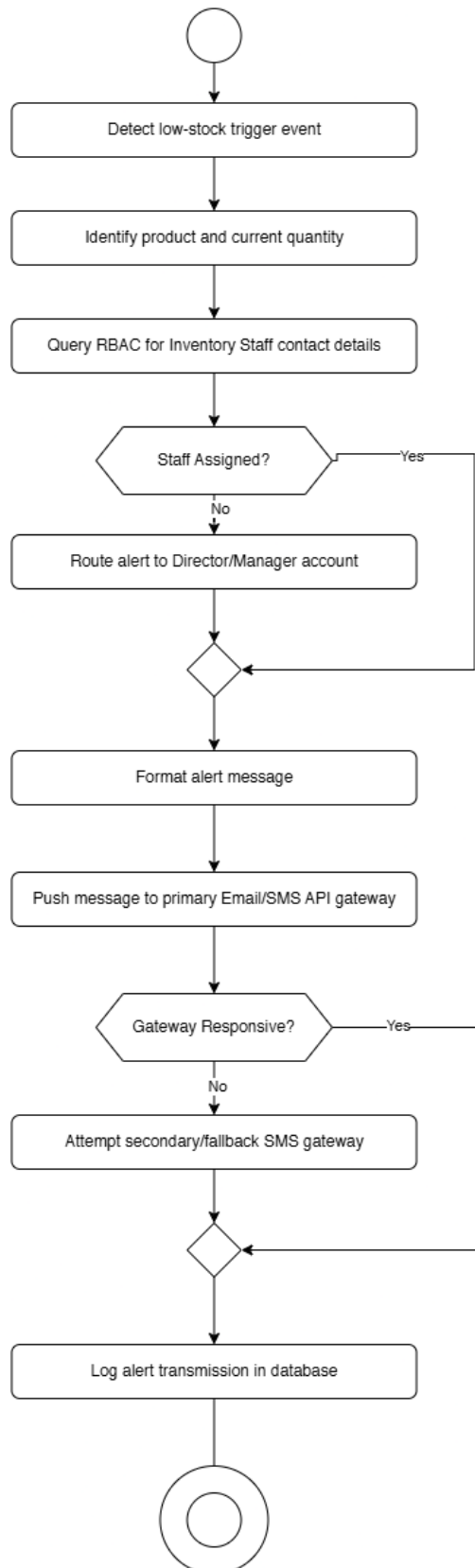
Activity Diagram: UC-02 Manage Stock Levels



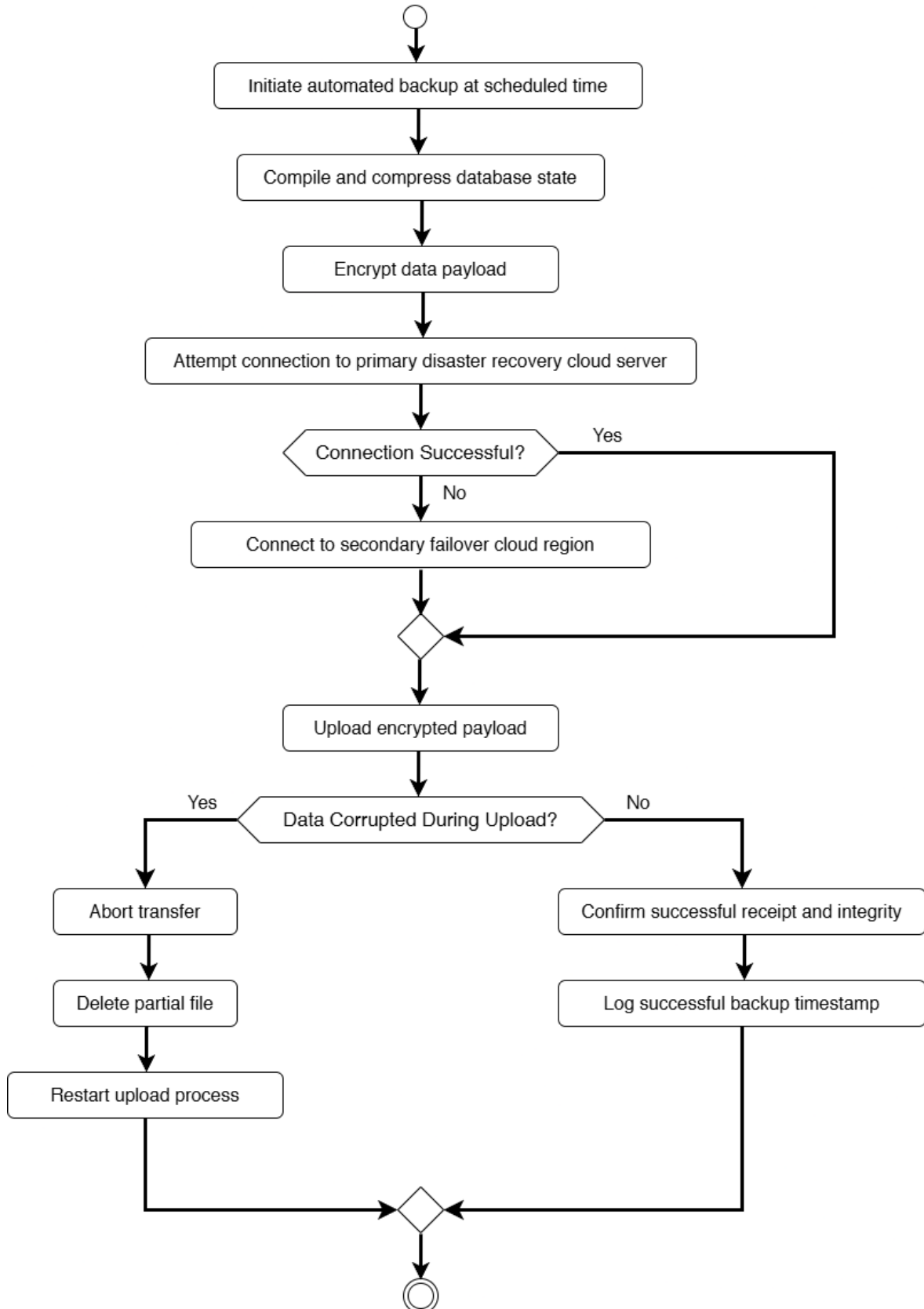
Activity Diagram: UC-03 Update Distribution Status



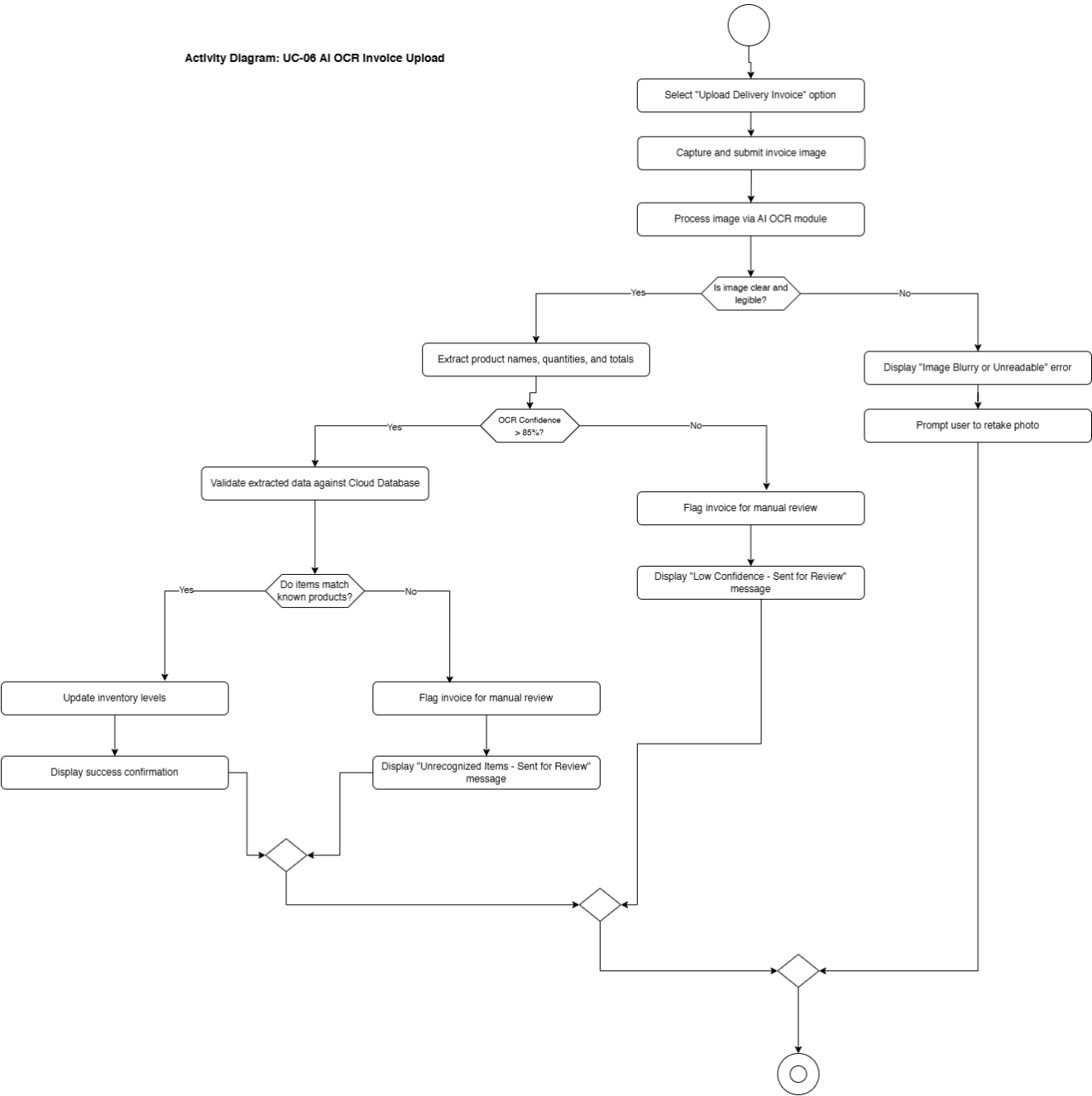
Activity Diagram: UC-04 Generate Low-Stock Alert



Activity Diagram: UC-05 Trigger Automated Cloud Backup

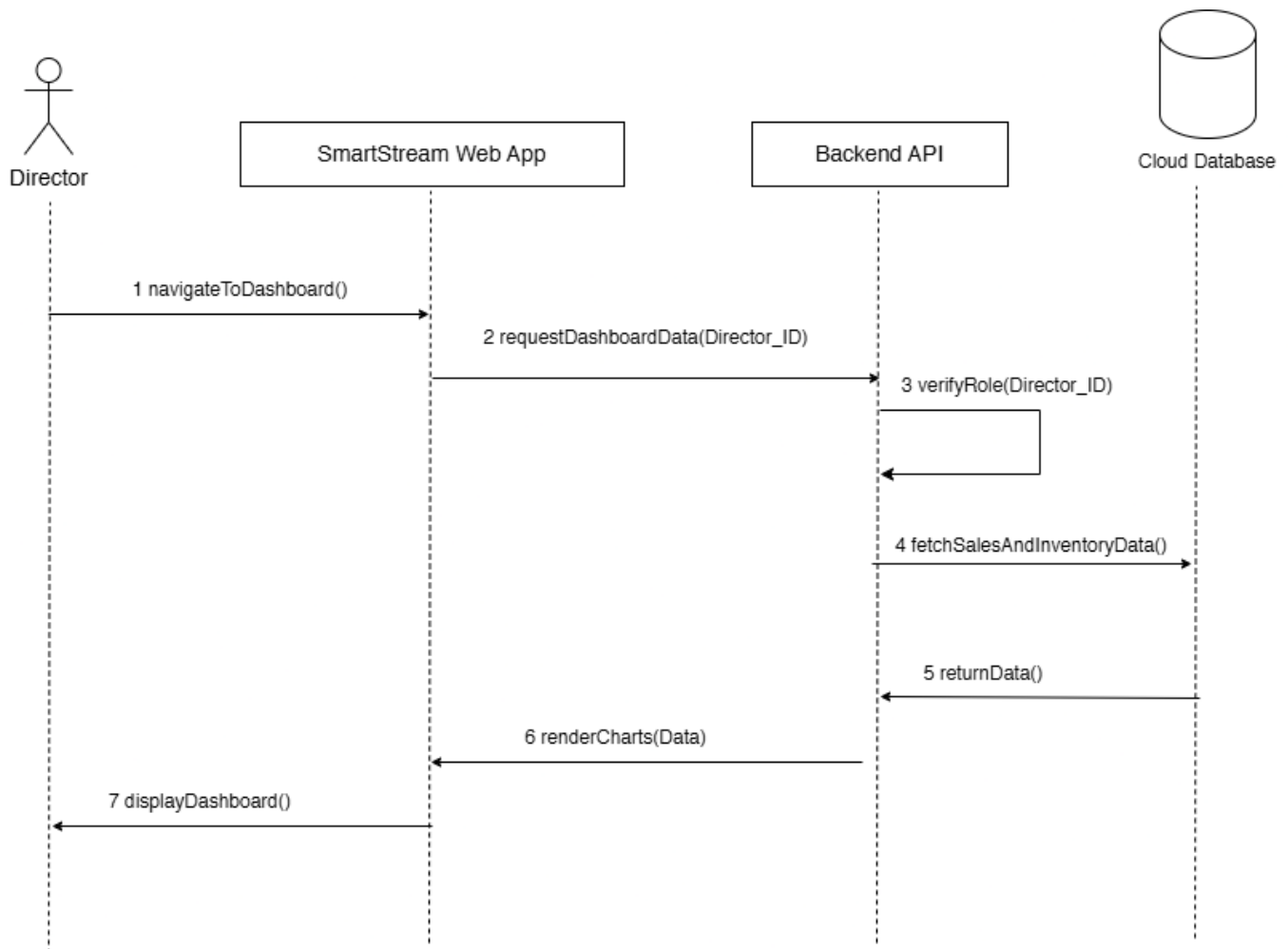


Activity Diagram: UC-08 AI OCR Invoice Upload

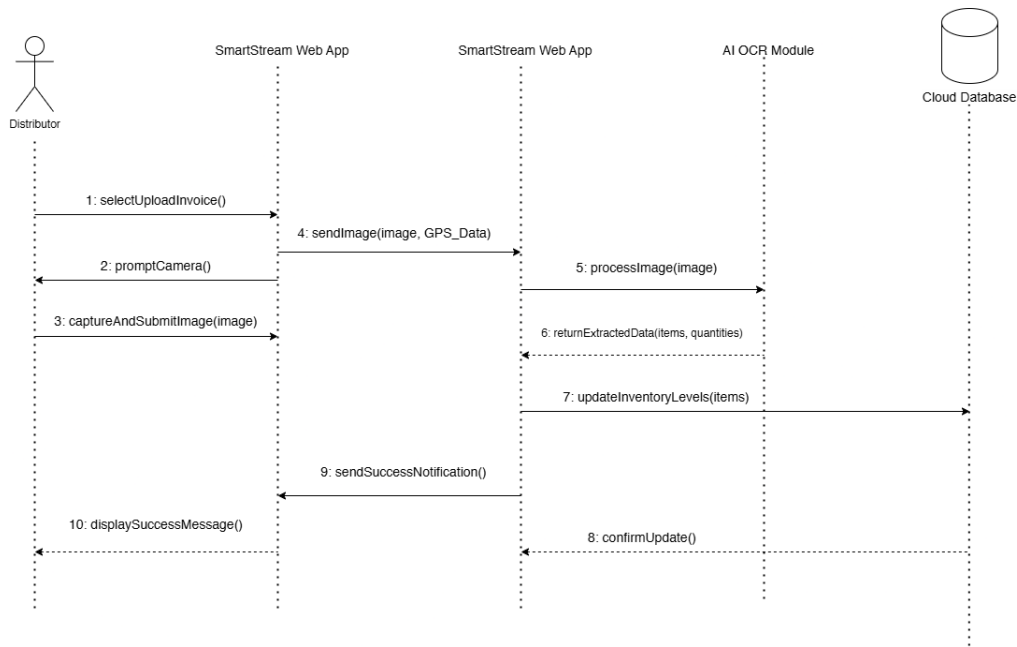


3.6 System Sequence Diagrams

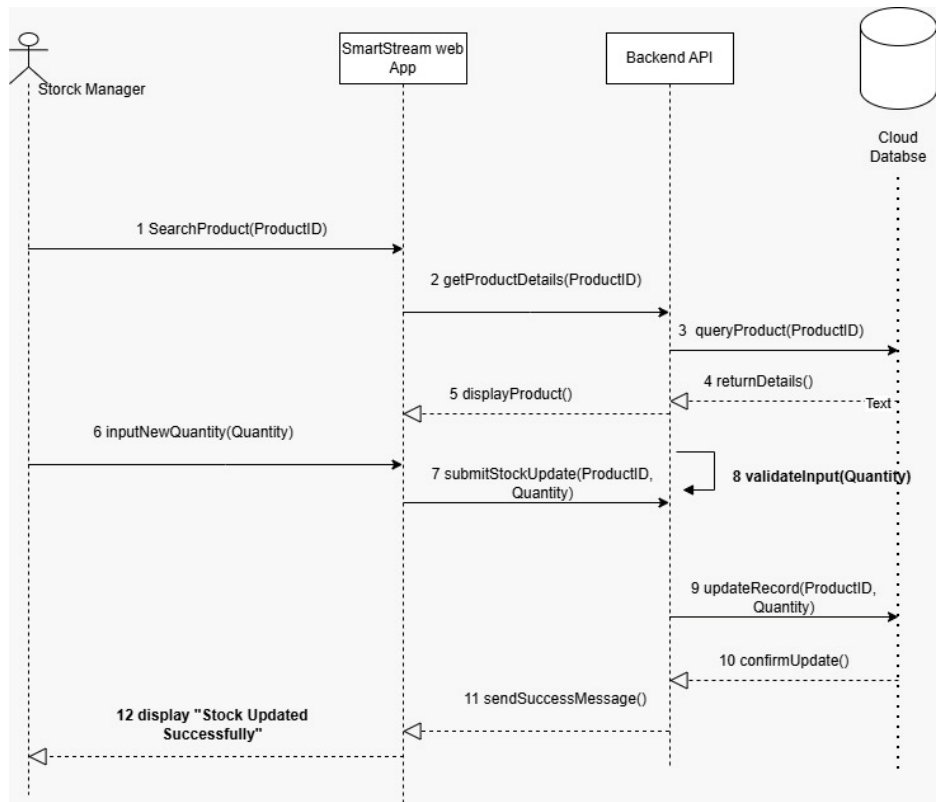
The following sequence diagrams illustrate the specific system interactions for each of the core user personas within SmartStream.



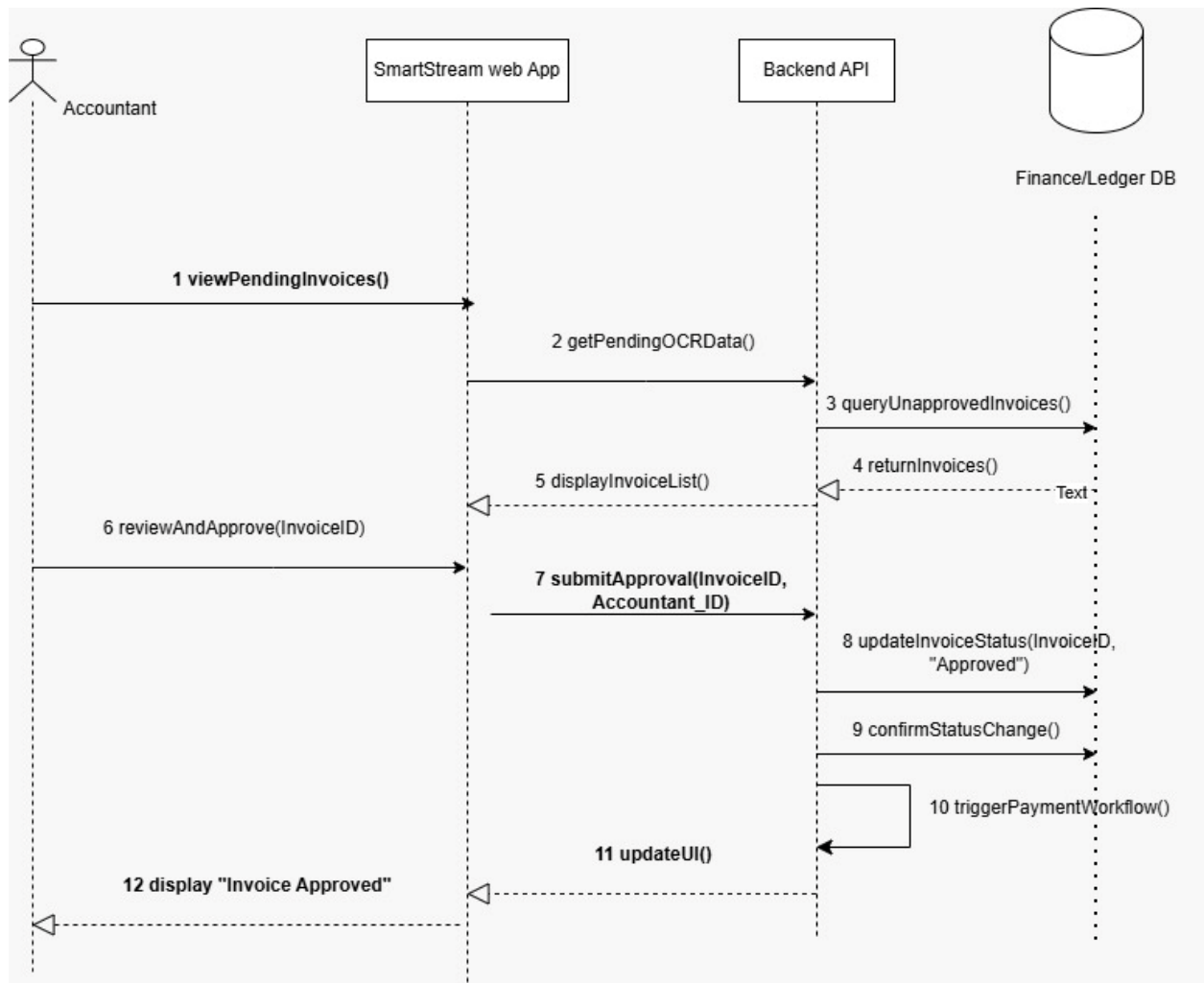
The Director: View Executive Dashboard



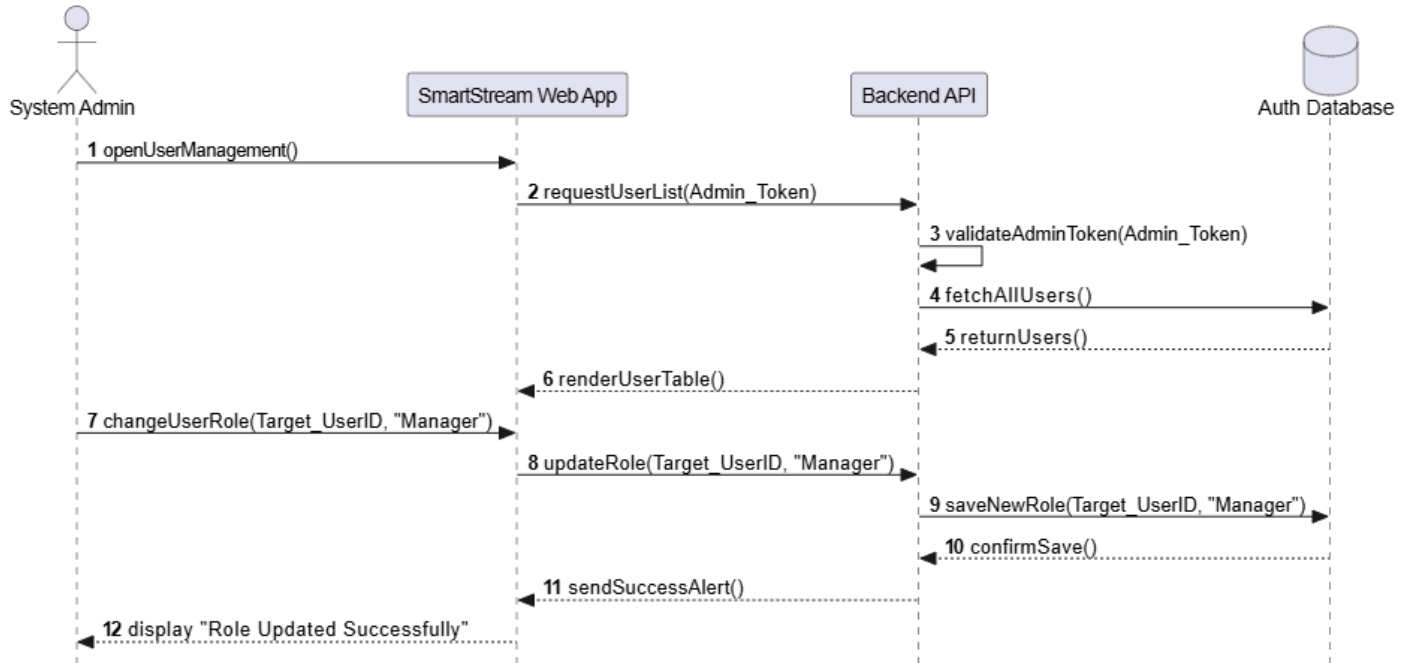
The Distributor: AI Invoice Upload



The Stock Manager: Manual Inventory Update

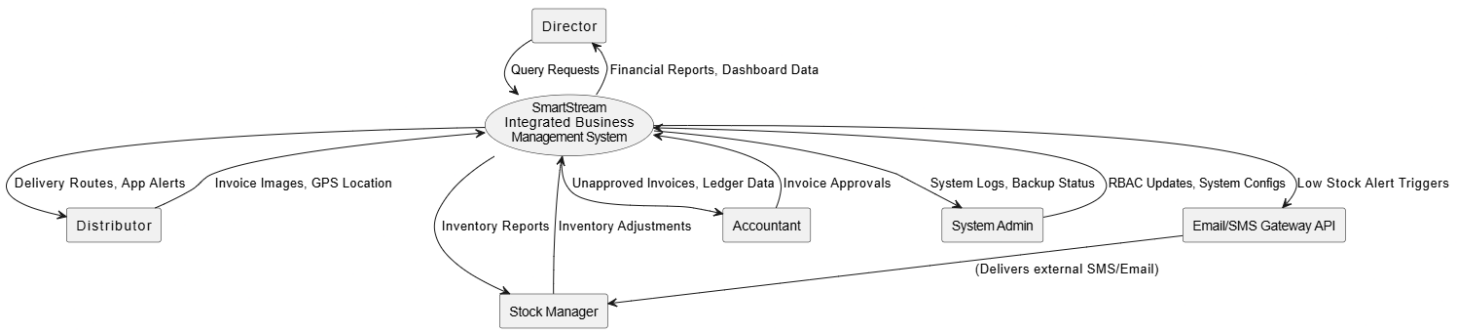


Accountant: Approval of Processed Invoices



The System Admin: Manage User Roles

3.7 System Context Diagram (Level 0 DFD)



4. Functional Requirements

ID	Title	Description	Priority	Source	Verification
FR-01	Role-Based Access Control	The system should restrict user access to modules and data based on assigned roles (Director, Manager, Distributor).	P1 – High	Client WhatsApp Request	Test Case TC-FR-01 (Security test)
FR-02	Automated Low Stock Alerts	The system shall automatically generate and send an email/SMS alert to the Stock Manager when a product's quantity falls below a defined threshold.	P1 – High	Client WhatsApp Request	Test Case TC-FR-02 (Integration test)
FR-03	AI OCR Data Extraction	The system shall extract	P2 – Medium	System Analysis /	Test Case TC-FR-03

ID	Title	Description	Priority	Source	Verification
		product names, quantities, and totals from uploaded invoice images using an AI OCR module.		Innovation	(Unit test)
FR-04	Real-time GPS Tracking	The system should track and log the real-time GPS coordinates of distributors while they are on an active delivery route.	P2 – Medium	Structured Interview	Test Case TC-FR-04 (System test)
FR-05	Executive Dashboard	The system shall provide a real-time dashboard for Directors displaying key	P1 – High	Client WhatsApp Request	Test Case TC-FR-05 (UI/UX test)

ID	Title	Description	Priority	Source	Verification
		performance indicators, total inventory value, and daily sales.			
FR-06	Cloud Disaster Recovery Backup	The system shall automatically backup the MySQL/PostgreSQL database to a secure cloud bucket every 12 hours.	P1 – High	Client WhatsApp Request	Test Case TC-FR-06 (DevOps test)

5. Non-Functional Requirements

- **Performance:** The AI OCR processing of an invoice image shall not exceed 5 seconds under standard network conditions. Dashboard analytics must load within 3 seconds.
- **Security:** All user passwords must be hashed using bcrypt. Data in transit must be encrypted via TLS 1.2 or higher. The system must adhere to standard data privacy practices.
- **Usability:** The mobile application UI must be accessible and intuitive, requiring no more than 3 taps for a distributor to upload an invoice.
- **Reliability:** The system shall maintain an uptime of 99.9%. In the event of a server failure, the cloud disaster recovery protocol must allow full system restoration within 4 hours.

- **Scalability:** The cloud backend (e.g., Node.js/Spring Boot) must support automatic scaling to accommodate up to 500 concurrent distributor requests.
- **Maintainability:** The system architecture will utilize a modular, decoupled approach (React frontend, separate REST API backend) to allow for isolated updates without system downtime.

6. Alternative Solutions & Feasibility Study

6.1 Analyzed Alternatives

Alternative 1: Commercial Off-The-Shelf Software (e.g., Zoho Inventory / Odoo)

- **Pros:** Fast implementation, proven reliability.
- **Cons:** Lacks highly specific custom features (like the specialized AI OCR pipeline and custom GPS tracking tailored exactly to KTD Lanka's distribution model). High recurring subscription costs.

Alternative 2: Custom On-Premises System

- **Pros:** Complete control over data, no recurring cloud hosting fees.
- **Cons:** Does not natively support the client's request for "remote access from any location" or robust "disaster recovery," requiring expensive in-house servers and IT staff.

6.2 Feasibility Analysis (Proposed Cloud-based SmartStream System)

- **Technical Feasibility (High):** The chosen tech stack (React, Flutter, Node.js, Cloud APIs) is modern, well-documented, and the development team possesses the necessary skills to integrate OCR and GPS tracking.
- **Economic Feasibility (Moderate to High):** While initial development costs are higher than an off-the-shelf subscription, the lack of per-user licensing fees will result in a positive ROI for KTD LANKA over a 3-year period.
- **Operational Feasibility (High):** The automated alerts and simplified mobile interface address the specific pain points of lost paperwork and manual entry, ensuring high user adoption rates from both staff and directors.
- **Schedule Feasibility (High):** The project scope is clearly defined and can be executed within standard agile sprints, aligning with university project deadlines.

6.3 Justified Recommendation

We recommend proceeding with the Custom Cloud-Based System (SmartStream). It perfectly

aligns with all client mandates from the requirements gathering phase—specifically the need for remote dashboard access, automated cloud disaster recovery, and specialized role-based tracking—which off-the-shelf and on-premises solutions fail to provide simultaneously and cost-effectively.